

PPM Framework
Hand-Solved Test Cases
Grouped by Mechanic
Internal Technical Reference
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Contents

Conventions

All cases share these conventions unless explicitly overridden:

- $\gamma = 1.0$ (no discounting), to keep arithmetic clean.
- $B_1 = 3 \cdot \sum_i \text{BAC}_i$ (large; cash never binding unless stated).
- $\bar{\eta}_{i,t} = \bar{\eta}$ constant and deterministic unless stated.
- Equal milestone weights $\phi_{i,j} = 1/M_i$ unless stated.
- $\alpha_i = 0, \rho_i = 0$ (no advance, no retention) unless stated.
- $\delta^{\text{rec}} = 0.25, \psi = 0.10$ (FIDIC defaults) unless stated.
- $\pi_i = 0$ unless stated ($CP_i = \text{BAC}_i$).
- S-curve shape $a_i = b_i = 2$ (symmetric Beta) unless stated.
- Monetary units are abstract.

1 Group 1: Progress and Milestone Certification

What is being proven: budget converts to progress at rate η , milestones certify at the correct threshold, payments trigger correctly, and milestone ordering is enforced even when progress jumps past multiple thresholds in a single period.

1.1 SP-1: Perfect Efficiency, Two Milestones, Front-Load

Economic claim. With $\eta = 1$, no budget pressure, and $\gamma = 1$, the MILP front-loads spend to certify milestones as early as possible. No termination conditions fire.

Parameters.

$$n = 1, \quad s_1 = 1, \quad f_1 = 4, \quad \text{BAC}_1 = 100, \quad CP_1 = 120, \quad \pi_1 = 0.20.$$

$$M_1 = 2, \quad \theta = (0.5, 1.0), \quad \phi = (0.5, 0.5), \quad \bar{\eta} = 1.0.$$

$$\alpha_1 = 0, \quad \rho_1 = 0, \quad \Omega_1 = 1, \quad \mu_1 = 0.30, \quad \tau_1^{\text{tol}} = 2.$$

$$B_1 = 300, \quad H = 4.$$

Milestone payments.

$$R_{1,1}^{\text{net}} = 0.5 \times 120 = 60, \quad R_{1,2}^{\text{net}} = 0.5 \times 120 = 60.$$

Identity: $0 + 60 + 60 + 0 = 120 = CP_1$. ✓

Optimal strategy. Spend 50 at $t = 1$ (MS1 certified), then 50 at $t = 2$ (MS2 certified).

t	$x_{1,t}$	$P_{1,t}$	Inflow	Net	B_t
1	50	0.50	60	+10	$300 \rightarrow 310$
2	50	1.00	60	+10	$310 \rightarrow 320$
3	0	1.00	0	0	320
4	0	1.00	0	0	320

Objective. $Z_{L1}^* = (60 - 50) + (60 - 50) = \mathbf{20}$.

Termination check. $\text{CPI} = 1.0$, $\text{EAC} = 100 \leq 1.30 \times 100$. Neither condition fires. ✓

Assertions for pytest.

A1. $x_{1,1} = 50 \pm 0.5$, $x_{1,2} = 50 \pm 0.5$, $x_{1,3} = x_{1,4} = 0$.

A2. $P_{1,1} = 0.50 \pm 0.01$, $P_{1,2} = 1.00 \pm 0.01$.

A3. MS1 certified at $t = 1$; MS2 certified at $t = 2$.

A4. Status: `completed`. No τ decrement.

A5. $|Z^* - 20| \leq 1.0$.

1.2 SP-3: Three Milestones, Uniform Allocation, $\gamma = 1$

Economic claim. With $\gamma = 1$, total net cash flow is independent of allocation timing within the feasible set. Any completing allocation achieves Z^* .

Parameters.

$$n = 1, \quad s_1 = 1, \quad f_1 = 6, \quad \text{BAC}_1 = 60, \quad \pi_1 = 0.50 \Rightarrow CP_1 = 90.$$

$$M_1 = 3, \quad \theta = (1/3, 2/3, 1), \quad \phi = (1/3, 1/3, 1/3), \quad \bar{\eta} = 1.0.$$

$$\alpha_1 = 0, \quad \rho_1 = 0, \quad \Omega_1 = 2, \quad \mu_1 = 0.50, \quad \tau_1^{\text{tol}} = 2.$$

Milestone payments. $R_{1,j}^{\text{net}} = 30$ per milestone. Total inflow = 90.

Uniform allocation. Spend $x_t = 10$ per period, milestones certify at $t = 2, 4, 6$. $Z^* = 90 - 60 = \mathbf{30}$.

Assertions for pytest.

A1. Total outflow = 60 ± 0.5 .

A2. All three milestones certified.

A3. Status: `completed`.

A4. $|Z^* - 30| \leq 1.0$.

A5. No termination conditions simultaneously active.

1.3 SP-J: Milestone Ordering Enforced on Progress Jump

Economic claim. When a single large allocation causes progress to cross multiple milestone thresholds in one period, the model must certify them in order. Both payments arrive in the same period but the ordering constraint $u_{i,j+1,t} \leq u_{i,j,t}$ is satisfied.

Parameters.

$$\begin{aligned}
 n = 1, \quad s_1 = 1, \quad f_1 = 4, \quad \text{BAC}_1 = 100, \quad CP_1 = 100, \quad \pi_1 = 0. \\
 M_1 = 2, \quad \theta = (0.5, 1.0), \quad \phi = (0.5, 0.5), \quad \bar{\eta} = 1.0. \\
 \alpha_1 = 0, \quad \rho_1 = 0, \quad \Omega_1 = 2, \quad \mu_1 = 0.50, \quad \tau_1^{\text{tol}} = 2. \\
 B_1 = 300, \quad H = 4.
 \end{aligned}$$

Allocation. Spend $x_{1,1} = 100$ at $t = 1$. Progress: $P_{1,1} = 1.0$, crossing both $\theta = 0.5$ and $\theta = 1.0$ simultaneously.

Payments at $t = 1$. MS1 certifies: $R_{1,1}^{\text{net}} = 50$. MS2 certifies: $R_{1,2}^{\text{net}} = 50$. Total inflow at $t = 1$: 100. Total outflow: 100. Net: 0.

Objective. $Z^* = 0$.

Assertions for pytest.

- A1. Both MS1 and MS2 certified at $t = 1$.
- A2. Ordering satisfied: MS1 certified before or simultaneously with MS2.
- A3. $P_{1,1} = 1.00 \pm 0.01$.
- A4. Status: `completed` by $t = 1$.
- A5. $|Z^* - 0| \leq 1.0$.

2 Group 2: Discounting and Timing Incentives

What is being proven: $\gamma < 1$ creates a genuine preference for earlier milestone certification; the MILP exploits this by front-loading allocations to projects whose milestone structure yields higher discounted inflows when completed first.

2.1 SP-4: Front-Loading Beats Uniform Allocation ($\gamma < 1$)

Economic claim. With discounting, earlier inflows are worth more. The MILP front-loads all spending to certify milestones at $t = 1$ and $t = 2$, beating the uniform allocation by a computable margin.

Parameters.

$$\begin{aligned}
 n = 1, \quad s_1 = 1, \quad f_1 = 8, \quad \text{BAC}_1 = 100, \quad \pi_1 = 0.40 \Rightarrow CP_1 = 140. \\
 M_1 = 2, \quad \theta = (0.5, 1), \quad \phi = (0.5, 0.5), \quad \bar{\eta} = 1.0, \quad \gamma = 0.95. \\
 \alpha_1 = 0, \quad \rho_1 = 0, \quad R_{1,j}^{\text{net}} = 70 \text{ each.}
 \end{aligned}$$

Front-loaded vs. uniform.

$$Z^{\text{front}} = 70 + 0.95 \times 70 - 50 - 0.95 \times 50 = 70 + 66.50 - 50 - 47.50 = \mathbf{39.00}.$$

Uniform (MS1 at $t = 4$, MS2 at $t = 8$):

$$Z^{\text{uniform}} = 0.95^3 \times 70 + 0.95^7 \times 70 - 12.5 \sum_{t=0}^7 0.95^t \approx 60.02 + 48.88 - 80.29 = \mathbf{28.61}.$$

Assertions for pytest.

- A1. $x_{1,1} = 50 \pm 0.5$, $x_{1,2} = 50 \pm 0.5$.
- A2. MS1 certified at $t = 1$; MS2 certified at $t = 2$.
- A3. $Z^* \geq 38.5$.
- A4. $Z^* > 28.61$.

2.2 MP-K: Two Projects, $\gamma < 1$, Front-Heavy vs. Back-Heavy Milestones

Economic claim. Two projects are identical in BAC, CP , η , and duration but differ in milestone weight distribution. Under $\gamma < 1$, the MILP completes the front-heavy project (large early payment) before the back-heavy project, since completing it first yields higher discounted total inflow.

Parameters.

$$n = 2, \quad s_i = 1, \quad f_i = 6, \quad \text{BAC}_i = 60, \quad \pi_i = 0.50 \Rightarrow CP_i = 90.$$

$$M_i = 2, \quad \theta_i = (0.5, 1.0), \quad \bar{\eta}_i = 1.0, \quad \gamma = 0.90.$$

$$\phi_A = (0.70, 0.30), \quad \phi_B = (0.30, 0.70).$$

$$\alpha_i = 0, \quad \rho_i = 0, \quad B_1 = 300.$$

Milestone payments. Project A: $R_{A,1}^{\text{net}} = 63$, $R_{A,2}^{\text{net}} = 27$. Project B: $R_{B,1}^{\text{net}} = 27$, $R_{B,2}^{\text{net}} = 63$. Each milestone requires spend of 30.

Strategy comparison. *Strategy 1 (A first, then B):*

$$Z = (63 - 30) + 0.90(27 - 30) + 0.81(27 - 30) + 0.729(63 - 30) = 33 - 2.7 - 2.43 + 24.06 = \mathbf{51.93}.$$

Strategy 2 (B first, then A):

$$Z = (27 - 30) + 0.90(63 - 30) + 0.81(63 - 30) + 0.729(27 - 30) = -3 + 29.7 + 26.73 - 2.19 = \mathbf{51.24}.$$

Strategy 1 dominates. MILP must choose A first or any strategy with $Z \geq 51.93$.

Assertions for pytest.

- A1. Project A completed before or simultaneously with Project B.
- A2. $Z^* \geq 51.90$.
- A3. $Z^* > 51.24$ (B-first strategy strictly dominated).
- A4. Both projects completed.

3 Group 3: Advance Payment and Recovery Mechanics

What is being proven: advance payment arrives at $t = s_i$, recovery deductions apply correctly once the cumulative payment threshold ψ_i is crossed, deductions stop once the advance is fully recovered, and the payment identity $A_i + \sum_j R_{i,j}^{\text{net}} + R_i^{\text{ret}} = CP_i$ holds exactly.

3.1 SP-2: Advance Harvest, Efficiency Decay, Forced Termination

Economic claim. A large advance combined with rapidly decaying efficiency makes the project economically dead within two periods. The MILP receives the advance and accepts termination. The settlement exactly cancels the advance, yielding $Z^* = 0$.

Parameters.

$$n = 1, \quad s_1 = 1, \quad f_1 = 6, \quad \text{BAC}_1 = 100, \quad CP_1 = 100, \quad \pi_1 = 0.$$

$$M_1 = 2, \quad \theta = (0.5, 1.0), \quad \phi = (0.5, 0.5).$$

$$\alpha_1 = 0.30 \Rightarrow A_1 = 30, \quad \delta^{\text{rec}} = 0.25, \quad \psi = 0.10, \quad \rho_1 = 0.$$

$$\Omega_1 = 1, \quad \mu_1 = 0.05, \quad \tau_1^{\text{tol}} = 1.$$

$$\bar{\eta}_{1,1} = 1.00, \quad \bar{\eta}_{1,t} = 0.05 \text{ for } t \geq 2.$$

Termination and settlement. With zero spend from $t = 1$: counter $1 \rightarrow 0$ at $t = 1$, termination at $t = 2$. $P_{1,2} = 0$. $R_1^{\text{term}} = 0 - 30 = -30$. $Z^* = 30 + 0 - 30 = 0$.

Assertions for pytest.

A1. $\sum_t x_{1,t} \leq 1.0$.

A2. Project terminates at $t \leq 3$.

A3. $R_1^{\text{term}} = -30 \pm 1$.

A4. No milestone certified.

A5. $|Z^* - 0| \leq 1.5$.

3.2 SP-A: Late Recovery Threshold ($\psi = 0.40$)

Economic claim. When $\psi = 0.40$, the cumulative certified payment fraction must exceed 40% before recovery deductions begin. MS1 carries weight $\phi_1 = 1/3 < 0.40$, so its payment is received in full. Deductions apply from MS2 onward. The payment identity holds exactly.

Parameters.

$$n = 1, \quad s_1 = 1, \quad f_1 = 6, \quad \text{BAC}_1 = 120, \quad \pi_1 = 0.50 \Rightarrow CP_1 = 180.$$

$$M_1 = 3, \quad \theta = (1/3, 2/3, 1), \quad \phi = (1/3, 1/3, 1/3).$$

$$\alpha_1 = 0.20 \Rightarrow A_1 = 36, \quad \psi = 0.40, \quad \delta^{\text{rec}} = 0.30, \quad \rho_1 = 0, \quad \bar{\eta} = 1.0.$$

$$\Omega_1 = 2, \quad \mu_1 = 0.50, \quad \tau_1^{\text{tol}} = 2, \quad B_1 = 400.$$

Recovery threshold check.

After MS1: $\sum_{k \leq 1} \phi_k = 1/3 \approx 0.333 < 0.40 \Rightarrow$ no deduction.

After MS2: $\sum_{k \leq 2} \phi_k = 2/3 \approx 0.667 \geq 0.40 \Rightarrow$ deduction applies.

After MS3: $\sum_{k \leq 3} \phi_k = 1.0 \geq 0.40 \Rightarrow$ deduction applies.

Net payments. Each milestone gross = 60.

$$\begin{aligned} R_{1,1}^{\text{net}} &= 60 - 0 = 60. \\ R_{1,2}^{\text{net}} &= 60 - 0.30 \times 60 = 60 - 18 = 42. \\ R_{1,3}^{\text{net}} &= 60 - 0.30 \times 60 = 60 - 18 = 42. \end{aligned}$$

Total deductions = 36 = A_1 . Identity: $36 + 60 + 42 + 42 = 180 = CP_1$. ✓

Allocation. Spend $x_t = 40$ per period; MS j certifies at $t = j$.

Objective. $Z^* = (36 + 60 - 40) + (42 - 40) + (42 - 40) = 56 + 2 + 2 = \mathbf{60}$.

Note: MS1 payment is received in full (no deduction) because cumulative weight $1/3 < \psi = 0.40$. This isolates the threshold mechanic from the SP-2 case where deductions begin at the first milestone.

Assertions for pytest.

- A1. $R_{1,1}^{\text{net}} = 60 \pm 0.5$ (no deduction).
- A2. $R_{1,2}^{\text{net}} = 42 \pm 0.5$ (deduction applies).
- A3. $R_{1,3}^{\text{net}} = 42 \pm 0.5$ (deduction applies).
- A4. Total inflow = 180 = CP_1 .
- A5. $|Z^* - 60| \leq 1.5$.

3.3 SP-B: Advance Fully Recovered Before Final Milestone

Economic claim. When recovery deductions on MS1 and MS2 exactly exhaust the advance, MS3 carries no deduction despite $\delta^{\text{rec}} > 0$. The final payment equals the gross milestone amount.

Parameters.

$$\begin{aligned} n &= 1, \quad s_1 = 1, \quad f_1 = 4, \quad \text{BAC}_1 = 60, \quad \pi_1 = 0.50 \Rightarrow CP_1 = 90. \\ M_1 &= 3, \quad \theta = (1/3, 2/3, 1), \quad \phi = (1/3, 1/3, 1/3). \\ \alpha_1 &= 0.20 \Rightarrow A_1 = 18, \quad \psi = 0.10, \quad \delta^{\text{rec}} = 0.30, \quad \rho_1 = 0, \quad \bar{\eta} = 1.0. \\ \Omega_1 &= 2, \quad \mu_1 = 0.50, \quad \tau_1^{\text{tol}} = 2, \quad B_1 = 200. \end{aligned}$$

Deduction schedule. Each milestone gross = 30. $\psi = 0.10$: all milestones eligible. Deduction per milestone = $0.30 \times 30 = 9$.

$$\begin{aligned} R_{1,1}^{\text{net}} &= 30 - 9 = 21. \quad \text{Recovered: 9.} \\ R_{1,2}^{\text{net}} &= 30 - 9 = 21. \quad \text{Recovered: 18 = } A_1. \quad \text{Advance fully recovered.} \\ R_{1,3}^{\text{net}} &= 30 - 0 = 30. \quad \text{No further deduction.} \end{aligned}$$

Identity: $18 + 21 + 21 + 30 = 90 = CP_1$. ✓

Objective. Spend $x_t = 20$ per period; MS j certifies at $t = j$. $Z^* = (18 + 21 - 20) + (21 - 20) + (30 - 20) = 19 + 1 + 10 = \mathbf{30}$.

Assertions for pytest.

- A1. $R_{1,1}^{\text{net}} = 21 \pm 0.5$.
- A2. $R_{1,2}^{\text{net}} = 21 \pm 0.5$.
- A3. $R_{1,3}^{\text{net}} = 30 \pm 0.5$ (no deduction on final milestone).
- A4. Total inflow = 90 = CP_1 .
- A5. $|Z^* - 30| \leq 1.0$.

4 Group 4: Retention

What is being proven: retention fraction ρ_i is withheld from every milestone payment, accumulates to $\rho_i \cdot CP_i$, and is released as a single lump sum at project completion only. Retention and advance recovery interact correctly when both are active.

4.1 SP-C: Retention Only, Lump Release at Completion

Economic claim. With $\rho = 0.10$ and no advance, each milestone payment is reduced by 10%. The withheld amounts accumulate and are released in full when the final milestone is certified.

Parameters.

$$\begin{aligned}
 n = 1, \quad s_1 = 1, \quad f_1 = 6, \quad \text{BAC}_1 = 60, \quad \pi_1 = 0.50 &\Rightarrow CP_1 = 90. \\
 M_1 = 3, \quad \theta = (1/3, 2/3, 1), \quad \phi = (1/3, 1/3, 1/3), \quad \bar{\eta} = 1.0. \\
 \alpha_1 = 0, \quad \rho_1 = 0.10, \quad \Omega_1 = 2, \quad \mu_1 = 0.50, \quad \tau_1^{\text{tol}} = 2.
 \end{aligned}$$

Net payments. Each gross milestone = 30. Retention per milestone = $0.10 \times 30 = 3$.

$$R_{1,j}^{\text{net}} = 30 \times (1 - 0.10) = 27 \quad \text{for } j = 1, 2, 3.$$

Retention release at completion: $R_1^{\text{ret}} = 0.10 \times 90 = 9$.

Final period inflow = $R_{1,3}^{\text{net}} + R_1^{\text{ret}} = 27 + 9 = 36$.

Identity: $0 + 27 + 27 + 27 + 9 = 90 = CP_1$. ✓

Objective. Spend $x_t = 20$ per period. $Z^* = (27 + 27 + 36) - (20 + 20 + 20) = 90 - 60 = \mathbf{30}$.

Assertions for pytest.

- A1. $R_{1,j}^{\text{net}} = 27 \pm 0.5$ for $j = 1, 2, 3$.
- A2. $R_1^{\text{ret}} = 9 \pm 0.5$ released at final certification only.
- A3. No retention released before project completed.
- A4. Total inflow = 90 = CP_1 .
- A5. $|Z^* - 30| \leq 1.0$.

4.2 SP-D: Retention and Advance Recovery Simultaneously Active

Economic claim. When both $\rho > 0$ and $\alpha > 0$, each milestone payment is reduced by both retention withholding and advance recovery deduction. After the advance is fully recovered, only retention applies. The payment identity holds exactly.

Parameters.

$$\begin{aligned}
n = 1, \quad s_1 = 1, \quad f_1 = 6, \quad \text{BAC}_1 = 60, \quad \pi_1 = 0.50 &\Rightarrow CP_1 = 90. \\
M_1 = 3, \quad \theta = (1/3, 2/3, 1), \quad \phi = (1/3, 1/3, 1/3), \quad \bar{\eta} = 1.0. \\
\alpha_1 = 0.20 &\Rightarrow A_1 = 18, \quad \psi = 0.10, \quad \delta^{\text{rec}} = 0.30, \quad \rho_1 = 0.10. \\
\Omega_1 = 2, \quad \mu_1 = 0.50, \quad \tau_1^{\text{tol}} = 2, \quad B_1 = 200.
\end{aligned}$$

Payment mechanics. Each gross milestone = 30. Retention per milestone = 3.

$$\begin{aligned}
R_{1,1}^{\text{net}} &= 30 \times 0.90 - 0.30 \times 30 = 27 - 9 = 18. \quad \text{Recovered: 9.} \\
R_{1,2}^{\text{net}} &= 27 - 9 = 18. \quad \text{Recovered: 18} = A_1. \quad \text{Advance fully recovered.} \\
R_{1,3}^{\text{net}} &= 27 - 0 = 27. \quad \text{Only retention; no recovery deduction.}
\end{aligned}$$

$$R_1^{\text{ret}} = 0.10 \times 90 = 9 \text{ at completion.}$$

$$\text{Identity: } 18 + 18 + 18 + 27 + 9 = 90 = CP_1. \quad \checkmark$$

Objective. $Z^* = (18 + 18 + 18 + 27 + 9) - (20 + 20 + 20) = 90 - 60 = \mathbf{30}$.

Assertions for pytest.

- A1. $R_{1,1}^{\text{net}} = 18 \pm 0.5$ (both retention and recovery deducted).
- A2. $R_{1,2}^{\text{net}} = 18 \pm 0.5$ (both deducted; advance now recovered).
- A3. $R_{1,3}^{\text{net}} = 27 \pm 0.5$ (retention only; no recovery deduction).
- A4. $R_1^{\text{ret}} = 9 \pm 0.5$ at completion.
- A5. Total inflow = 90 = CP_1 .
- A6. $|Z^* - 30| \leq 1.0$.

5 Group 5: Termination Conditions in Isolation

What is being proven: Conditions 1 and 2 are independent. The cure-period counter decrements only when *both* conditions hold simultaneously. When only one holds, the counter must remain at τ_i^{tol} . When one lapses mid-cure, the counter resets.

Note: A key identity: $\text{CPI}_{i,t} = \eta_{i,t}$ always. Therefore $\text{CPI} = 1$ if and only if $\eta = 1$, and $\text{EAC} = \text{BAC}$ whenever $\eta = 1$, making Condition 2 structurally impossible to fire under $\eta = 1$.

5.1 SP-E: Only Condition 1 Fires (Schedule Lag, $\text{CPI} = 1$)

Economic claim. With $\eta = 1$, $\text{CPI} = 1$ always and $\text{EAC} = \text{BAC}$, so Condition 2 cannot fire regardless of spend. Spending far below the planned rate causes $\text{SPI} < 1$ and Condition 1 to fire. The counter must stay at τ^{tol} throughout.

Parameters.

$$\begin{aligned}
n = 1, \quad s_1 = 1, \quad f_1 = 8, \quad \text{BAC}_1 = 80, \quad CP_1 = 80, \quad \pi_1 = 0. \\
M_1 = 1, \quad \theta = (1.0), \quad \phi = (1.0), \quad \bar{\eta} = 1.0. \\
\alpha_1 = 0, \quad \rho_1 = 0, \quad \Omega_1 = 1, \quad \mu_1 = 0.30, \quad \tau_1^{\text{tol}} = 3. \\
a_1 = b_1 = 1 \text{ (uniform S-curve),} \quad B_1 = 300.
\end{aligned}$$

Spend $x_t = 5$ per period (half the uniform plan rate of 10).

Condition checks at $t = 3$. $P_3^{\text{actual}} = 15/80 = 0.1875$. $P_3^{\text{plan}} = (3-1)/(8-1) = 2/7 \approx 0.286$. $\text{SPI} = 0.1875/0.286 = 0.656 < 1$. Condition 1 fires. $\text{CPI} = \eta = 1.0$. $\text{EAC} = 15 + (80 - 15)/1 = 80 \leq 1.30 \times 80$. Condition 2 *does not* fire.

Counter at every period: $\tau^{\text{rem}} = 3$ (unchanged).

Assertions for pytest.

- A1. $\text{SPI}_{1,t} < 1$ for $t \geq 3$.
- A2. $\text{CPI}_{1,t} = 1.0 \pm 0.01$ for all t .
- A3. $\text{EAC}_{1,t} \leq (1 + \mu_1)\text{BAC}_1$ for all t .
- A4. $\tau_{1,t}^{\text{rem}} = 3$ for all t (counter never decrements).
- A5. Status never **terminated**.

5.2 SP-F: Only Condition 2 Fires (Cost Overrun, $\text{SPI} \geq 1$)

Economic claim. With $\eta = 0.5$, $\text{CPI} = 0.5$ and EAC quickly exceeds $(1 + \mu)\text{BAC}$. A very long planned duration and large spending keeps $\text{SPI} \geq 1$, so Condition 1 does not fire. The counter must stay at τ^{tol} .

Parameters.

$$n = 1, \quad s_1 = 1, \quad f_1 = 20, \quad \text{BAC}_1 = 100, \quad \text{CP}_1 = 100, \quad \pi_1 = 0.$$

$$M_1 = 1, \quad \theta = (1.0), \quad \phi = (1.0), \quad \bar{\eta} = 0.5.$$

$$\alpha_1 = 0, \quad \rho_1 = 0, \quad \Omega_1 = 5, \quad \mu_1 = 0.05, \quad \tau_1^{\text{tol}} = 3.$$

$$a_1 = b_1 = 1 \text{ (uniform S-curve)}, \quad B_1 = 500.$$

Spend $x_t = 10$ per period.

Condition checks at $t = 3$. $P_3^{\text{actual}} = 0.5 \times 30/100 = 0.15$. $P_3^{\text{plan}} = (3-1)/(20-1) = 2/19 \approx 0.105$. $\text{SPI} = 0.15/0.105 = 1.43 > 1$. Ahead of schedule. Condition 1 *does not* fire. $\text{CPI} = 0.5$. $\text{EAC} = 30 + (100 - 15)/0.5 = 30 + 170 = 200 > 1.05 \times 100 = 105$. Condition 2 fires.

Counter at every period: $\tau^{\text{rem}} = 3$ (unchanged).

Assertions for pytest.

- A1. $\text{SPI}_{1,t} > 1$ for $t \leq 10$.
- A2. $\text{EAC}_{1,t} > 105$ from $t = 1$ onward.
- A3. $\tau_{1,t}^{\text{rem}} = 3$ for all observed periods (counter never decrements).
- A4. Status never **terminated**.

5.3 SP-G: Both Fire, Then Condition 1 Lapses — Counter Resets

Economic claim. After the counter has decremented twice under simultaneous conditions, a large allocation in one period recovers the schedule sufficiently to lapse Condition 1. The counter must reset to τ^{tol} immediately.

Parameters.

$$n = 1, \quad s_1 = 1, \quad f_1 = 12, \quad \text{BAC}_1 = 100, \quad CP_1 = 100, \quad \pi_1 = 0.$$

$$M_1 = 1, \quad \theta = (1.0), \quad \phi = (1.0).$$

$$\alpha_1 = 0, \quad \rho_1 = 0, \quad \Omega_1 = 0, \quad \mu_1 = 0.50, \quad \tau_1^{\text{tol}} = 3.$$

$$a_1 = b_1 = 1 \text{ (uniform S-curve)}, \quad B_1 = 500.$$

Efficiency schedule: $\bar{\eta}_t = 0.3$ for $t = 1, 2$; $\bar{\eta}_3 = 1.0$. Spend $x_t = 10$ for $t = 1, 2$; $x_3 = 80$.

Period-by-period trace. $t = 1$: $P_1^{\text{actual}} = 0.03$. $P_1^{\text{plan}} = 0$. Convention: $\text{SPI} = 1$ when $\text{plan} = 0$. Only CPI fires ($\text{CPI} = 0.3$, $\text{EAC} = 10 + 233 = 243 > 150$). Counter: stays at 3.

$t = 2$: $P_2^{\text{actual}} = 0.06$. $P_2^{\text{plan}} = 1/11 \approx 0.091$. $\text{SPI} = 0.06/0.091 = 0.66 < 1$. Δf fires. Both conditions active. Counter: $3 \rightarrow 2$.

$t = 3$: $P_3^{\text{actual}} = 0.06 + 1.0 \times 80/100 = 0.86$. $P_3^{\text{plan}} = 2/11 \approx 0.182$. $\text{SPI} = 0.86/0.182 = 4.7 > 1$. $\hat{f}_3 = 3 + (12 - 3 + 1 - 1)/4.7 = 3 + 1.91 = 4.91 < 12$. $\Delta f < 0 \leq \Omega = 0$. Condition 1 lapses. Counter resets to 3.

Assertions for pytest.

- A1. $\tau_{1,1}^{\text{rem}} = 3$ (only one condition at $t = 1$).
- A2. $\tau_{1,2}^{\text{rem}} = 2$ (both conditions; decrement).
- A3. $\tau_{1,3}^{\text{rem}} = 3$ (Condition 1 lapses; counter resets).
- A4. Status at $t = 3$: **active** (not terminated).

6 Group 6: Cure Period and Termination Execution

What is being proven: the counter decrements correctly period by period, termination executes exactly in the period after the counter reaches zero, the settlement is computed at the correct progress level, and the counter restart mechanic is correctly accounted for when computing the total time to termination.

6.1 SP-5: Zero Efficiency, Exact Counter Boundary

Economic claim. With $\eta = 0$, no progress is achievable. Both conditions fire from the start. With $\tau^{\text{tol}} = 3$, termination executes at $t = 4$. The settlement exactly cancels the advance, yielding $Z^* = 0$.

Parameters.

$$n = 1, \quad s_1 = 1, \quad f_1 = 6, \quad \text{BAC}_1 = 100, \quad CP_1 = 100, \quad \pi_1 = 0.$$

$$M_1 = 1, \quad \theta = (1.0), \quad \phi = (1.0).$$

$$\alpha_1 = 0.20 \Rightarrow A_1 = 20, \quad \delta^{\text{rec}} = 0.25, \quad \psi = 0.10, \quad \rho_1 = 0.$$

$$\tau_1^{\text{tol}} = 3, \quad \Omega_1 = 1, \quad \mu_1 = 0.10, \quad \bar{\eta} = 0.$$

Termination timeline.

t	Both conditions?	τ^{rem}	Event
1	Yes	$3 \rightarrow 2$	Decrement
2	Yes	$2 \rightarrow 1$	Decrement
3	Yes	$1 \rightarrow 0$	Counter hits zero
4	—	—	Termination executes

$$P_{1,4} = 0. \quad R_1^{\text{term}} = 0 - (20 - 0) = -20. \quad Z^* = 20 - 20 = \mathbf{0}.$$

Assertions for pytest.

$$\text{A1. } \tau_{1,1}^{\text{rem}} = 2, \tau_{1,2}^{\text{rem}} = 1, \tau_{1,3}^{\text{rem}} = 0.$$

$$\text{A2. Termination period} = 4.$$

$$\text{A3. } R_1^{\text{term}} = -20 \pm 0.5.$$

$$\text{A4. Status: } \texttt{terminated}.$$

$$\text{A5. } |Z^* - 0| \leq 1.0.$$

6.2 SP-H: Partial Progress at Termination, Positive Settlement

Economic claim. A project that reaches partial progress before conditions fire produces a positive termination settlement when the value of work in place exceeds the outstanding advance obligation. The MILP prefers this outcome over zero spend.

Parameters.

$$n = 1, \quad s_1 = 1, \quad f_1 = 10, \quad \text{BAC}_1 = 100, \quad CP_1 = 100, \quad \pi_1 = 0.$$

$$M_1 = 2, \quad \theta = (0.30, 1.0), \quad \phi = (0.40, 0.60).$$

$$\alpha_1 = 0.10 \Rightarrow A_1 = 10, \quad \psi = 0.10, \quad \delta^{\text{rec}} = 0.25, \quad \rho_1 = 0.$$

$$\bar{\eta} = 0.5, \quad \Omega_1 = 0, \quad \mu_1 = 0.05, \quad \tau_1^{\text{tol}} = 2.$$

$$a_1 = 0.5, \quad b_1 = 2 \text{ (front-loaded plan)}, \quad B_1 = 300.$$

Strategy: spend 60 at $t = 1$. $P_{1,1} = 0.5 \times 60/100 = 0.30$. MS1 certified ($\theta_1 = 0.30$): $R_{1,1}^{\text{net}} = 0.40 \times 100 - 0.25 \times 0.40 \times 100 = 40 - 10 = 30$. (Deduction applies since $\phi_1 = 0.40 \geq \psi = 0.10$.) Advance recovered: 10. Outstanding advance: 0.

Condition checks at $t = 1$: $P_1^{\text{plan}} = I_{0.1}(0.5, 2) \approx 0.47$. $\text{SPI} = 0.30/0.47 = 0.638 < 1$. Condition 1 fires. $\text{CPI} = 0.5$. $\text{EAC} = 60 + (100 - 30)/0.5 = 200 > 1.05 \times 100$. Condition 2 fires. Counter: $2 \rightarrow 1$.

No spend at $t = 2$. Both conditions persist. Counter: $1 \rightarrow 0$. Termination executes at $t = 3$. $P_{1,3} = 0.30$.

Settlement. Outstanding advance after MS1 deduction = 0.

$$R_1^{\text{term}} = 0.30 \times 100 \times (1 - 0) - 0 = 30 > 0. \quad \checkmark$$

Cash flows.

t	x	Inflow	Net	B_t
1	60	$A_1 = 10, R_{1,1}^{\text{net}} = 30$	-20	$300 \rightarrow 280$
2	0	0	0	280
3	0	$R_1^{\text{term}} = 30$	+30	310

$$Z^* = (10 + 30 - 60) + 0 + 30 = \mathbf{10}.$$

Note: Zero-spend strategy: $P = 0$, $R_1^{\text{term}} = -10$, $Z = 10 - 10 = 0 < 10$. MILP correctly prefers spending 60 at $t = 1$.

Assertions for pytest.

A1. MS1 certified at $t = 1$; $R_{1,1}^{\text{net}} = 30 \pm 0.5$.

A2. $\tau_{1,1}^{\text{rem}} = 1$, $\tau_{1,2}^{\text{rem}} = 0$.

A3. Termination at $t = 3$.

A4. $R_1^{\text{term}} = 30 \pm 0.5$ (positive).

A5. $|Z^* - 10| \leq 1.5$.

6.3 SP-I: Cure Period Interrupted and Restarted

Economic claim. After the counter has decremented, a large mid-cure allocation that restores schedule causes the counter to reset. Subsequent renewed deterioration restarts the full cure period. The total time to termination is therefore longer than the naïve count of τ^{tol} periods.

Parameters.

$$n = 1, \quad s_1 = 1, \quad f_1 = 12, \quad \text{BAC}_1 = 100, \quad \text{CP}_1 = 100, \quad \pi_1 = 0.$$

$$M_1 = 1, \quad \theta = (1.0), \quad \phi = (1.0).$$

$$\alpha_1 = 0, \quad \rho_1 = 0, \quad \Omega_1 = 0, \quad \mu_1 = 0.50, \quad \tau_1^{\text{tol}} = 3.$$

$$a_1 = b_1 = 1 \text{ (uniform)}, \quad B_1 = 500.$$

Efficiency schedule: $\bar{\eta}_t = 0.3$ for $t = 1, 2$; $\bar{\eta}_3 = 1.0$, $x_3 = 80$ (recovery spike); $\bar{\eta}_t = 0.3$ for $t \geq 4$, $x_t = 10$.

Counter trajectory.

t	Both conditions?	τ^{rem}	Reason
1	No (plan= 0)	3	Only Cond. 2; no decrement
2	Yes	$3 \rightarrow 2$	Both fire
3	No (Cond. 1 lapses after spike)	$2 \rightarrow 3$	Reset to τ^{tol}
4	Yes (slow spend resumes)	$3 \rightarrow 2$	Both fire
5	Yes	$2 \rightarrow 1$	Both fire
6	Yes	$1 \rightarrow 0$	Counter hits zero
7	—	—	Termination executes

Without the recovery spike, termination would execute at $t = 4$ (counter $3 \rightarrow 2 \rightarrow 1 \rightarrow 0$ over $t = 1, 2, 3$). With the reset, it executes at $t = 7$ — three additional periods later.

Assertions for pytest.

- A1. $\tau_{1,2}^{\text{rem}} = 2$ (decrement at $t = 2$).
- A2. $\tau_{1,3}^{\text{rem}} = 3$ (reset at $t = 3$ after Cond. 1 lapses).
- A3. $\tau_{1,6}^{\text{rem}} = 0$.
- A4. Termination period = 7 (not $t = 4$).
- A5. Status: `terminated`.

7 Group 7: Portfolio Cash Balance and Coupling

What is being proven: B_t is the sole coupling between projects; inflows from one project correctly fund allocations to another; the no-borrowing constraint is respected; and negative settlement payments reduce the shared balance and constrain subsequent allocations.

7.1 MP-3: Advance Harvest Funds Concurrent Good Project

Economic claim. The advance from the Bad project, received at $t = 1$, enters the shared balance and is immediately available to fund the Good project. The net result is identical whether or not the advance exists, because the settlement exactly cancels it.

Parameters. Same as MP-2 (below) but $B_1 = 90 = 0.5 \times (100 + 80)$.

Period-by-period trace.

t	x_G	x_B	Inflow	Net	B_t
1	40	0	$A_B = 30, R_{G,1} = 60$	+50	$90 \rightarrow 140$
2	40	0	$R_{G,2} = 60$	+20	$140 \rightarrow 160$
3	0	0	$R_B^{\text{term}} = -30$	-30	$160 \rightarrow 130$

$Z^* = 50 + 20 - 30 = \mathbf{40}$. $B_t \geq 0$ throughout. ✓

Assertions for pytest.

- A1. $B_t \geq 0$ for all t .
- A2. Bad terminates; advance received then cancelled by settlement.
- A3. Good: `completed` by $t = 2$.
- A4. $|Z^* - 40| \leq 1.5$.

7.2 MP-A: Sequential Funding — Second Project Waits for Milestone Inflow

Economic claim. When B_1 covers only the first milestone allocation of Project 1, Project 2 cannot receive any allocation until Project 1's first milestone payment replenishes the shared balance.

Parameters.

$n = 2$. Both: $s_i = 1$, $f_i = 6$, $BAC_i = 60$, $\pi_i = 0.50 \Rightarrow CP_i = 90$.

$M_i = 2$, $\theta_i = (0.5, 1.0)$, $\phi_i = (0.5, 0.5)$, $\bar{\eta}_i = 1.0$.

$\alpha_i = 0$, $\rho_i = 0$, $\Omega_i = 2$, $\mu_i = 0.50$, $\tau_i^{\text{tol}} = 2$.

$B_1 = 30$ (covers exactly one first-milestone spend of P1).

Trace. $t = 1$: $B_1 = 30$. Spend 30 on P1 ($P_{1,1} = 0.5$, MS1 certified, $R_{1,1}^{\text{net}} = 45$). Cannot allocate to P2 (B exhausted before inflow). $B_2 = 30 - 30 + 45 = 45$.

$t = 2$: $B_2 = 45$. Spend 30 on P1 (MS2, $R_{1,2}^{\text{net}} = 45$). Spend 15 on P2 ($P_{2,2} = 0.25$, MS1 not yet certified). $B_3 = 45 - 45 + 45 = 45$.

$t = 3$: Spend 15 on P2 ($P_{2,3} = 0.5$, MS1 certified, $R_{2,1}^{\text{net}} = 45$). $B_4 = 45 - 15 + 45 = 75$.

$t = 4$: Spend 30 on P2 (MS2, $R_{2,2}^{\text{net}} = 45$). $B_5 = 75 - 30 + 45 = 90$.

$Z^* = (30 - 30) + (45 + 45 - 30 - 15) + (45 - 15) + (45 - 30) \dots$

Let me compute directly: total inflow = $45 + 45 + 45 + 45 = 180$. Total outflow = $30 + 30 + 15 + 15 + 30 = 120$. $Z^* = 180 - 120 = \mathbf{60}$.

Key assertion: $x_{2,1} = 0$ (P2 receives no allocation at $t = 1$ because B_1 is exhausted funding P1's first milestone).

Assertions for pytest.

A1. $x_{2,1} = 0$ (no allocation to P2 at $t = 1$).

A2. $B_t \geq 0$ for all t .

A3. Both projects completed.

A4. $|Z^* - 60| \leq 2.0$.

7.3 MP-B: Negative Settlement Drains Shared Balance

Economic claim. When a terminated project produces a negative settlement $R^{\text{term}} < 0$, the shared balance B_t is reduced accordingly. Subsequent allocations to surviving projects must respect the reduced balance.

Parameters.

$n = 2$.

Project Term: $s = 1$, $f = 8$, $BAC = 60$, $CP = 60$, $\pi = 0$, $M = 1$, $\theta = (1.0)$, $\phi = (1.0)$, $\alpha = 0.30 \Rightarrow A = 18$, $\psi = 0.10$, $\delta^{\text{rec}} = 0.25$, $\rho = 0$, $\bar{\eta} = 0$, $\Omega = 1$, $\mu = 0.10$, $\tau^{\text{tol}} = 2$.

Project Good: $s = 1$, $f = 8$, $BAC = 60$, $\pi = 0.50 \Rightarrow CP = 90$, $M = 2$, $\theta = (0.5, 1.0)$, $\phi = (0.5, 0.5)$, $\alpha = 0$, $\rho = 0$, $\bar{\eta} = 1.0$, $\Omega = 2$, $\mu = 0.50$, $\tau^{\text{tol}} = 3$.

$B_1 = 100$.

Trace. $t = 1$: Receive $A_{\text{Term}} = 18$. Spend 30 on Good. $P_{\text{Good},1} = 0.5$. MS1_{Good} certified: $R_{\text{Good},1} = 45$. $B_2 = 100 + 18 - 30 + 45 = 133$. Term: $\eta = 0$, both conditions fire, counter $2 \rightarrow 1$.

$t = 2$: Term counter $1 \rightarrow 0$. Spend 30 on Good. $P_{\text{Good},2} = 1.0$. MS2_{Good} certified: $R_{\text{Good},2} = 45$. $B_3 = 133 - 30 + 45 = 148$.

$t = 3$: Term executes. $P_{\text{Term}} = 0$. $R_{\text{Term}}^{\text{term}} = 0 - (18 - 0) = -18$. $B_4 = 148 + (-18) = 130$.

After $t = 3$: Good is completed; no further allocations needed. $Z^* = (18 - 30 + 45) + (45 - 30) + (-18) = 33 + 15 - 18 = \mathbf{30}$.

Key assertion: $B_4 = 130$ (reduced by $|R^{\text{term}}| = 18$ relative to $B_3 = 148$); no constraint violation.

Assertions for pytest.

- A1. $R_{\text{Term}}^{\text{term}} = -18 \pm 0.5$.
- A2. $B_4 = 130 \pm 1$ (balance reduced by settlement).
- A3. $B_t \geq 0$ for all t .
- A4. Good: `completed`. Term: `terminated`.
- A5. $|Z^* - 30| \leq 1.5$.

8 Group 8: Portfolio Selection Under Constraints

What is being proven: when capital is scarce, the MILP correctly prioritises higher-NPV projects, may starve lower-NPV projects to termination, and evaluates the option value of funding a low-NPV project whose early milestone provides cash that unlocks completion of a higher-NPV project.

8.1 MP-2: Good vs. Bad Project, Unconstrained Cash

Economic claim. With abundant cash, the MILP allocates zero to the Bad project (which terminates yielding net zero) and maximises spend on the Good project. $Z^* = Z_{\text{Good}}$.

Parameters. *Project Bad:* $\text{BAC} = 100$, $CP = 100$, $\pi = 0$, $s = f^{\text{plan}} = 1 \dots 8$, $M = 2$, $\theta = (0.5, 1)$, $\alpha = 0.30$ ($A = 30$), $\delta^{\text{rec}} = 0.25$, $\psi = 0.10$, $\rho = 0$, $\Omega = 1$, $\mu = 0.10$, $\tau^{\text{tol}} = 2$. $\bar{\eta}_1 = 1.0$; $\bar{\eta}_t = 0.05$ for $t \geq 2$.

Project Good: $\text{BAC} = 80$, $CP = 120$, $\pi = 0.50$, $s = 1$, $f = 8$, $M = 2$, $\theta = (0.5, 1)$, $\alpha = 0$, $\rho = 0$, $\Omega = 2$, $\mu = 0.30$, $\tau^{\text{tol}} = 3$, $\bar{\eta} = 1.0$.

$H = 8$. $B_1 = 540$.

Bad terminates at $t = 4$; net contribution = 0. Good: front-load $x = 40$ at $t = 1, 2$; $Z_G = (60 - 40) + (60 - 40) = 40$. $Z^* = 40$.

Assertions for pytest.

- A1. $\sum_t x_{\text{Bad},t} \leq 2.0$.
- A2. Bad terminates at $t \leq 4$.
- A3. Good: `completed`.
- A4. $|Z^* - 40| \leq 1.5$.

8.2 MP-4: Three Projects, Low Always Terminates

Economic claim. Project Low has $\eta = 0.5$, making any spend immediately trigger an EAC overrun. The MILP terminates Low and completes High and Mid.

Parameters. $H = 6$, $B_1 = 150$, $\gamma = 1$. All: $\alpha = 0$, $\rho = 0$, $\tau^{\text{tol}} = 2$, $\delta^{\text{rec}} = 0.25$, $\psi = 0.10$.

Proj.	BAC	π	CP	$\bar{\eta}$	Ω/μ
High	60	0.50	90	1.0	2/0.40
Mid	60	0.20	72	1.0	2/0.30
Low	60	0.10	66	0.5	1/0.10

$$Z^* = Z_H + Z_M + Z_L = (45 - 30) + (36 - 30) + (45 - 30) + (36 - 30) + 0 = 15 + 6 + 15 + 6 = \mathbf{42}.$$

Assertions for pytest.

- A1. High and Mid **completed** by $t = 2$.
- A2. Low **terminated** by $t \leq 4$; allocation to Low ≤ 2 .
- A3. $|Z^* - 42| \leq 1.5$.
- A4. $B_t \geq 0$ for all t .

8.3 MP-C: Two Positive-NPV Projects, Cash Sufficient for Only One

Economic claim. B_1 is sufficient to complete exactly one project. The MILP completes the higher-margin project; the lower-margin project receives zero allocation and terminates, yielding net zero contribution.

Parameters.

$$n = 2, \quad s_i = 1, \quad f_i = 4.$$

Project Hi: $BAC = 60$, $\pi = 0.50 \Rightarrow CP = 90$, $M = 2$, $\theta = (0.5, 1)$, $\phi = (0.5, 0.5)$, $\bar{\eta} = 1.0$, $\alpha = 0$, $\rho = 0$, $\Omega = 2$, $\mu = 0.50$, $\tau^{\text{tol}} = 2$.

Project Lo: $BAC = 60$, $\pi = 0.10 \Rightarrow CP = 66$, $M = 2$, $\theta = (0.5, 1)$, $\phi = (0.5, 0.5)$, $\bar{\eta} = 1.0$, $\alpha = 0$, $\rho = 0$, $\Omega = 2$, $\mu = 0.50$, $\tau^{\text{tol}} = 2$.

$B_1 = 60$ (covers exactly BAC_{Hi} , insufficient for both).

Analysis. Completing Hi: outflow 60, inflow 90. $Z_{Hi} = 30$. Completing Lo: outflow 60, inflow 66. $Z_{Lo} = 6$. Cannot complete both simultaneously (would require $B_1 \geq 120$).

With $B_1 = 60$: MILP allocates fully to Hi. Lo receives zero, both conditions fire from $t = 1$ (with zero spend, SPI undefined; default to termination). Lo terminates; $A_{Lo} = 0$, $R_{Lo}^{\text{term}} = 0$.

$$Z^* = Z_{Hi} + 0 = \mathbf{30}.$$

Assertions for pytest.

- A1. $\sum_t x_{Lo,t} \leq 1.0$.
- A2. Lo: **terminated**.
- A3. Hi: **completed**.
- A4. $|Z^* - 30| \leq 1.5$.
- A5. $B_t \geq 0$ for all t .

8.4 MP-D: Early Milestone Cash Unlocks Higher-NPV Project

Economic claim. A low-NPV project with an early milestone can be worth funding partially if its payment replenishes the balance sufficiently to enable completion of a higher-NPV project that would otherwise stall. The MILP evaluates this option correctly.

Parameters.

$$n = 3, \quad s_i = 1, \quad B_1 = 40.$$

Project Bridge: $f = 3$, $BAC = 30$, $\pi = 0.10 \Rightarrow CP = 33$, $M = 1$, $\theta = (1.0)$, $\phi = (1.0)$, $\bar{\eta} = 1.0$, $\alpha = 0$, $\rho = 0$, $\Omega = 2$, $\mu = 0.50$, $\tau^{\text{tol}} = 2$.

Project Alpha: $f = 6$, $BAC = 60$, $\pi = 0.50 \Rightarrow CP = 90$, $M = 2$, $\theta = (0.5, 1)$, $\phi = (0.5, 0.5)$, $\bar{\eta} = 1.0$, $\alpha = 0$, $\rho = 0$, $\Omega = 2$, $\mu = 0.50$, $\tau^{\text{tol}} = 2$.

Project Beta: same as Alpha.

Strategy 1: fund Bridge first. $t = 1$: spend 30 on Bridge. $B_2 = 40 - 30 + 33 = 43$. $t = 2$: spend 30 on Alpha (MS1, $R = 45$), 13 on Beta. $B_3 = 43 - 43 + 45 = 45$. $t = 3$: spend 17 on Beta (cum. 30, MS1 $_{\beta}$, $R = 45$), 28 on Alpha. $B_4 = 45 - 45 + 45 = 45$. $t = 4$: spend 30 on Alpha (MS2, $R = 45$), 15 on Beta. $B_5 = 45 - 45 + 45 = 45$. $t = 5$: spend 15 on Beta (MS2, $R = 45$). $B_6 = 45 - 15 + 45 = 75$.

This gets complex. The key economic point is simpler to state: $Z_{\text{Bridge}} = 33 - 30 = 3$ (small positive). If Bridge enables completion of Alpha and Beta which otherwise stall, the total $Z = 3 + 30 + 30 = 63$ versus not funding Bridge and completing only one of Alpha/Beta with $Z = 30$. MILP must recognize the option value of Bridge.

Assertions for pytest.

- A1. Bridge funded at $t = 1$: $x_{\text{Bridge},1} \geq 28$.
- A2. Bridge completed.
- A3. $Z^* > 30$ (MILP does better than completing only one major project).
- A4. $B_t \geq 0$ for all t .

9 Group 9: Horizon and Staggered Starts

What is being proven: H is derived correctly as $\max_i f_i - \min_i s_i + 1$; allocations outside each project's active window $[s_i, f_i]$ are zero; EVM signals are only computed within active windows; non-overlapping projects have additive objectives.

9.1 MP-1: Two Non-Overlapping Projects, Additive Objective

Economic claim. Two projects with no temporal overlap behave independently. $Z^* = Z_A + Z_B$.

Parameters. Project A (SP-1): $s = 1$, $f = 4$. Project B (SP-3): $s = 5$, $f = 10$. $H = \max(4, 10) - \min(1, 5) + 1 = 10$.

$$Z^* = 20 + 30 = 50.$$

Assertions for pytest.

- A1. $|Z^* - 50| \leq 1.0$.
- A2. Both completed.
- A3. $x_{A,t} = 0$ for $t \geq 5$; $x_{B,t} = 0$ for $t \leq 4$.
- A4. No terminations.

9.2 MP-E: Three Projects with Staggered Starts

Economic claim. Projects starting at $s_i \in \{1, 4, 9\}$ produce $H = 14$. Allocations outside each project's active window are structurally zero. EVM signals are undefined (and should not be computed) before $t = s_i$.

Parameters.

$$n = 3.$$

Proj.	s	f	BAC	π	M	$\bar{\eta}$
P1	1	4	60	0.50	2	1.0
P2	4	8	60	0.50	2	1.0
P3	9	14	60	0.50	2	1.0

$H = \max(4, 8, 14) - \min(1, 4, 9) + 1 = 14 - 1 + 1 = 14$. All: $\alpha = 0$, $\rho = 0$, $\Omega = 2$, $\mu = 0.50$, $\tau^{\text{tol}} = 2$.

Each project: $\phi = (0.5, 0.5)$, $CP = 90$, $R_j^{\text{net}} = 45$ each. Outflow per project: 60. $Z_i = 90 - 60 = 30$ each. $Z^* = 3 \times 30 = \mathbf{90}$.

Assertions for pytest.

- A1. $H = 14$ (derived correctly).
- A2. $x_{1,t} = 0$ for $t \geq 5$.
- A3. $x_{2,t} = 0$ for $t \leq 3$ and $t \geq 9$.
- A4. $x_{3,t} = 0$ for $t \leq 8$.
- A5. No EVM signal computed before $t = s_i$ for each project.
- A6. All three **completed**.
- A7. $|Z^* - 90| \leq 2.0$.

10 Summary Table

Case	Group	Pattern tested	Outcome	Z^*	Key assertion
SP-1	G1	Perfect η , 2 MS, baseline	Completed	20	$x = [50, 50, 0, 0]$; both MS certified
SP-3	G1	3 MS, uniform, $\gamma = 1$	Completed	30	Outflow= 60; any timing gives same Z
SP-J	G1	Progress jump, ordering enforced	Completed	0	Both MS at $t = 1$; order satisfied
SP-4	G2	Front-load beats uniform, $\gamma < 1$	Completed	39	$Z^* > 28.61$; MS1 at $t = 1$
MP-K	G2	Front-heavy ϕ prioritised, $\gamma < 1$	Both comp.	51.93	A before B; $Z > 51.24$
SP-2	G3	Advance harvest, decay, termination	Terminated	0	$x \approx 0$; settlement= -30
SP-A	G3	Late threshold $\psi = 0.40$	Completed	60	MS1 full; MS2,3 deducted

Case	Group	Pattern tested	Outcome	Z^*	Key assertion
SP-B	G3	Advance recovered before final MS	Completed	30	MS3 full despite $\delta^{\text{rec}} > 0$
SP-C	G4	Retention only, lump release	Completed	30	Each MS= 27; release= 9 at end
SP-D	G4	Retention + advance together	Completed	30	MS1,2 both deducted; MS3 retention only
SP-E	G5	Only Cond. 1 fires ($\eta = 1$)	Active	—	Counter stays at τ^{tol}
SP-F	G5	Only Cond. 2 fires (long plan)	Active	—	Counter stays at τ^{tol}
SP-G	G5	Both fire; Cond. 1 lapses; reset	Active	—	$\tau = [3, 2, 3]$ at $t = 1, 2, 3$
SP-5	G6	$\eta = 0$, exact counter boundary	Terminated	0	$\tau = [2, 1, 0]$; term at $t = 4$
SP-H	G6	Partial progress, positive R^{term}	Terminated	10	$R^{\text{term}} = 30 > 0$; MILP spends
SP-I	G6	Cure reset; later termination	Terminated	—	Term at $t = 7$ not $t = 4$
MP-3	G7	Advance funds concurrent project	Mixed	40	$B_t \geq 0$; advance absorbed
MP-A	G7	Sequential funding, cash depleted	Both comp.	60	$x_{2,1} = 0$; P2 waits for MS1 inflow
MP-B	G7	Negative settlement drains balance	Mixed	30	$B_4 = 130$; balance reduced
MP-2	G8	Good vs. Bad, large B_1	Bad term.	40	Alloc to Bad ≈ 0
MP-4	G8	Low η forces termination	Low term.	42	H&M complete; Low terminates
MP-C	G8	Cash for one project only	Lo term.	30	Hi completed; Lo starved
MP-D	G8	Bridge cash unlocks major projects	All comp.	> 30	Bridge funded; option value realised
MP-1	G9	Non-overlapping, additive	Both comp.	50	$Z = Z_A + Z_B$; no interference
MP-E	G9	Staggered starts, H derived	All comp.	90	$H = 14$; zero alloc outside windows

Table 1: All test cases grouped by mechanic.